radmind Manual for Mac OS X

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With the help of the radmind development team and beta testers

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For latest version of this document go to
http://www.gal.co.uk/software/radmind/

Radmind was created by the Research Systems UNIX Group at The University of Michigan

For more information and latest release go to http://www.radmind.org/



1. The basic idea

radmind (**r**emote **admin**istration **d**aemon) is a client management system that allows you to create a specific setup on a single Mac OS X system and then be able to implement the same setup on multiple clients. Most importantly, radmind enables you to install updates and new apps on a single Mac and then force the other Macs to inherit the same configuration. It can be set to automatically bring back systems to a pristine state every night in a college lab or as a way to distribute new system updates on demand.

radmind supports multiple configurations so one radmind server can handle several departments in your organization, each with its own setup and applications.

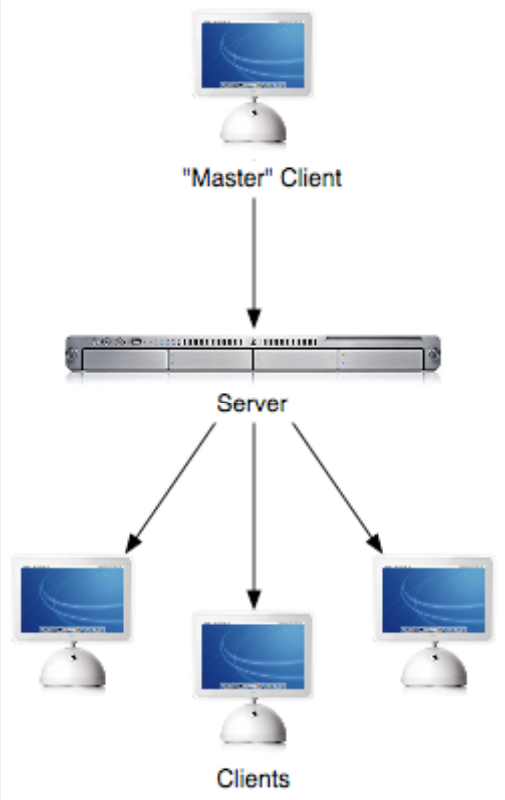
At its core, radmind operates as a tripwire; it is able to detect differences between the server and the client to any managed file system object, e.g. files, directories, links, etc. However, radmind goes further than just integrity checking: once a difference is detected, radmind can optionally take action.

This is ideal for small to large businesses as well as schools and universities. radmind not only lets you upgrade and keep all systems the same, it also lets you downgrade if you need to. radmind is generally useful if you have three or more Macs that need to run similar or identical configurations.

You can use radmind to combat any application or system corruption and even deliberate mis-configuration by simply running the radmind update session. When used with checksums, radmind also verifies the integrity of files and any damaged ones are replaced.

radmind even works in super user mode (Command+S) allowing a system administrator to repair a system that won't start properly.

radmind can be used in conjunction with Apple Software Restore (ASR), NetBoot, NetInstall and Carbon Copy Cloner.



The "master" client is used to upload the base OS and any additional applications, drivers and updates to the server.

Any client can then match the master by simply initiating a radmind update.

radmind is a very powerful tool that can also delete important files. It is therefore recommended that you read this document through and only then attempt to use radmind. It is also a good idea to experiment on test systems before deploying the setup in the real world. OS X, unlike its predecessors, installs a large number of files, and you'll want to choose which to manage in the process of your testing.

radmind does not require a special "master" client for generating updates for other clients; any radmind client can become the master by simply updating it with radmind.

radmind can be set to skip user data and other files – that is leave user documents untouched while updating the rest of the system.

radmind can be started manually, or automated to run at startup, login, logout or at timed intervals. But before you get ahead of yourself, let's start with the basics…

2. How radmind works

radmind uses a client-server setup. The server holds all the files required to make a client match a specified configuration. Such a configuration may include OS X and various applications used in your organization.

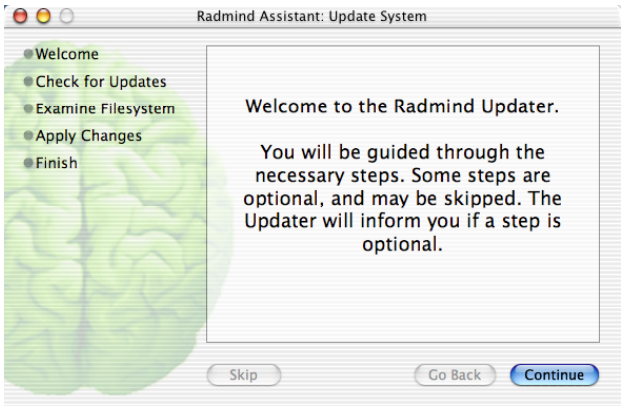


You can install the server component on any Mac running OSX 10.2 or later, or any UNIX/Linux based systems. It doesn't have to be an OSX Server. Most main functions of radmind, including server management, are available via the radmind Assistant application (Mac OS X only). If you prefer you may also use the Terminal to run radmind.

Initially, a client is used to create the base setup which is then uploaded to the server. When the setup is verified and saved on the server any other client can be configured to connect to the server and initiate a radmind update session – effectively downloading and installing all the files required to match the first client.

Normal client updates start by downloading the various radmind server files that describe the required setup for that client. radmind then scans the client's file system for any differences between the prescribed setup and the actual files on the client. If any mismatch is found you are then prompted to perform an update which will make the client match the prescribed setup.

In principal, to create a basic working setup you need to install OSX and all your standard applications on one workstation (the source client) and then use radmind Assistant to upload the setup to the server. Once there, any other client with radmind installed can initiate an update, which will result in making the new client an identical clone of the source client.



The radmind Assistant simplifies the radmind process and enables a 3-click client update.

You can then install additional applications on the source client and again upload the changes to the server. The other clients can then be updated to match the new setup.

Whenever radmind updates a client it deletes any files that were not on the source and copies any missing or modified files from the server. In some ways this works much like folder synchronizers such as psync or rsync. In other words it makes the target **identical** to the source as defined on the radmind server.

You can determine which files end up on the clients on the server by using a combination of setup files – the radmind preferences. By employing such a scheme radmind is able to deliver different files depending on which client is being updated. Most of these files can be configured using the Server Manager which is included with the radmind Assistant.

In order to skip files and folders that should not be touched with each update such as the Users folder, radmind uses the negative transcript. Example negative transcripts are included with radmind and also on the radmind.org site. These include user files as well as various logs and cache files that should normally be managed differently.

A finely tuned negative transcript is key to a successful radmind installation and consequent client updates. See the next section to learn more about how it works.

3. What are transcripts, loadsets and command files

Overview

radmind uses three file types to store information about client configuration. These are organised in a hierarchy that allows for a very flexible setup – capable of managing a diverse organisation where each department may require a different setup.

- The radmind server uses a single configuration file (config) to store a list of clients and their associated command files.
- Each command file contains a list of transcripts (.T)
- Each transcript contains a list of files and their attributes

More about the radmind file hierarchy

Each transcript describes the contents of a loadset which is simply the files & folders required to deploy the transcript. A transcript called MSOffice.T for example, may contain a list of all the files installed by Microsoft Office and the loadset will contain the actual Microsoft Office files and folders.

A command file is used to bring several transcripts and their corresponding Loadsets together. One client may have a command file that includes OS X, Microsoft Office and FileMaker Pro loadsets, while another client could have a command file with the same OSX loadset, scanner drivers and Adobe Photoshop.

You can easily assign each command file you have to many clients using wildcards and IP address ranges, allowing you to distribute different application sets to different departments for example.

When editing any radmind files by hand, make sure you use an editor capable of handling very long lines and Unix mode linefeeds. If you use pico remember to use –w option.

Transcripts (.T)

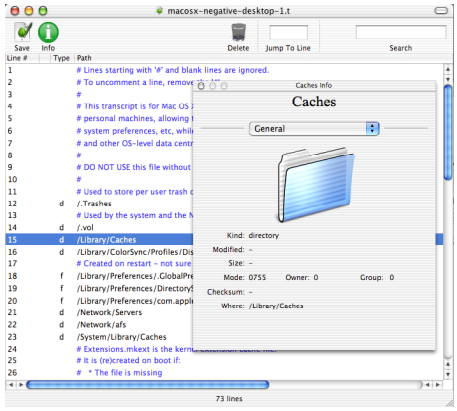
A transcript is a plain text file containing a list of files with instructions for radmind. A transcript for a base OS X installation contains 10,000s lines corresponding to each file system object (i.e. files, folders, etc.).

Each line in the transcript lists one system object – a file, folder, link, etc. In addition the privileges settings, the files size and modification date are also listed. The object type is denoted with a single letter where f for example stands for a regular file, d stands for a folder (directory) and h is a hard link. For a full list of object types consult the fsdiff man pages.

You may optionally include checksums to verify the integrity of files, but use of checksums is known to cause unnecessary file copying with the current version of OS X (10.2.x), which optimises some files on the fly. On the other hand, checksums may catch out some viruses and other system hacks.

When updating a client, radmind compares the state of the clients to the transcripts listed in its command file. It then produces a transcript that describes what changes need to be made to bring the client up-to-date. This temporary transcript is also referred to as apply-able since its contents will be applied to a client.

In an apply-able transcript a + in front of a file name means it is to be copied to the client, while a – indicates that the file will be deleted. If a file appears with no +/- its attributes need changing – most likely its privileges. If a managed file on a client is different it will be replaced, if it is missing radmind will copy it to the client. If a file on the client has no match in your transcripts it will be deleted. Similarly, if a file has different privileges these will also be adjusted to match your transcripts.



The Transcript Editor can be used to view and edit radmind transcripts.

This means that if a user installed an application or a printer driver they will be deleted by radmind, but you can use the negative transcript as well as other methods to work around this if needed.

radmind determines which files should be created, copied, deleted or modified based on your server configuration in where a command file is used to determine which transcripts to employ. The use of multiple transcripts allows greater flexibility and enables you, among other things, to add software and updates to clients quite easily.

Transcripts are normally created on a client and are then uploaded, along with their corresponding files, to the server where they can be made available to other clients.

Most radmind client updates start by downloading the transcripts to the client. This ensures that the client has an up-to-date version.

Note that the sort order of items in a transcript is crucial for smooth operation of radmind. The Transcript Editor ensures that your transcripts are properly sorted. If you plan on using a text editor instead, make sure you understand the sort order required.

Positive Transcripts

Most of your transcripts would be positive. A positive transcript contains a list of file system objects that should be added, modified or deleted. If a file has been modified on the client

(this is decided by date, size, privileges and optionally checksums) it would be replaced with the server version of that file.

Most non-user files should normally appear in a positive transcript. This includes the System, Applications, drivers, root Library, etc.

Negative Transcripts

If you use tools like rsync or Retrospect you will be familiar with the concept of an exclusion list. At first glance it may look like the negative transcript is just that – a list of files and folders that should not be managed. radmind doesn't work exactly like any of these tools.

The negative transcript is not just an exclusion list!

Instead, radmind uses the negative transcript to help you maintain a working system on your client and that can mean creating folders and files if they're missing, or in some cases just tweaking them.

This way radmind can ensure, for example, that each client has a Users folder in the right place with the correct permissions, but will not manage its contents. In addition to the Users folder, other items such as host name, netinfo database and various cache files would normally be found in a negative transcript.

In most cases, items in the negative transcripts are only managed in the sense that radmind ensures their existence and attributes, but does not manage their contents, whether the object is a file or a folder. If a file is not found on the client radmind will copy it across. This is in contrast with positive transcript items that are fully managed and any changes to their contents will trigger a radmind update.

When creating a new loadset, whether it's a base load or an overload, radmind client checks the negative transcript and excludes any items listed from the new loadset.

For better understanding of the negative transcript you should read the fsdiff man page, but in most cases understanding the description above will serve you well.

A group of radmind users is currently running an experimental web site that lists some OS X files and their purpose. It is available at http://gal.homeip.net:591/ – enter *guest* as both user name and password and click on FSObjects in the next page. This is a temporary location; always check the radmind.org web site for an up-to-date link.

Special Transcripts

To do

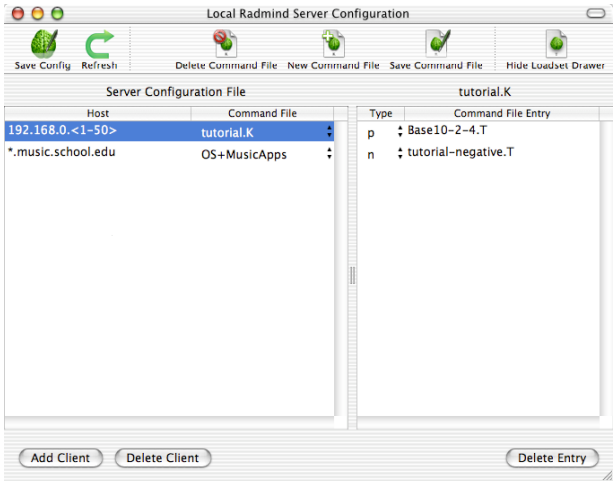
Command files (.K)

A command file is a plain text file containing a list of transcripts in an ascending order of priority from top to bottom, with the last item having the highest priority. A command file must contain at least one negative and one positive transcripts. The negative transcript should normally have the highest priority and should therefore be listed last.

You may use one transcript in several command files. This simplifies your setup and reduces your server storage requirements.

If a file appears in more than one transcript, the one lower in the command file takes precedence. This enables you to update clients from iMovie 3.0.1 to 3.0.2 for example, by placing the transcript that contains the 3.0.2 update lower in the command file.

The only exception to this rule is the negative transcript. Items that appear in the negative transcript should in most cases **not** appear in any of the positive transcripts. Normally radmind takes care of that, but if you modify the negative transcript, you must also ensure that all active positive transcripts reflect this change.



The server configuration window allows you to create, edit and assign command files to clients.

Using the Server Manager you can assign different command files to specific Macs using their IP address or name to identify them. By default only one host is setup to apply to all clients using the * wildcard. You may use a combination of IP ranges, wildcards and specific addresses to assign different command files to groups of machines. You may have one command file for your music class containing the OS and overloads with music apps, one for graphic design class with graphic tools and another for school staff. The same arrangement equally applies to a business with several departments. For some users a single global host (such as 192.168.1.<1-50>) will suffice, but it should be noted that leaving the wildcard entry means that anyone with access to your server will be able to download any files from the radmind server so you may want to use a firewall to prevent unauthorised access.

If you want to use more than one command file you must remove the default “*” host or at least make sure it’s listed last.

Your final hosts configuration, which is saved in the server config file, may look something like this:

192.168.0.<1-10>	servers.K
192.168.0.<11-20>	admins.K
192.168.0.<21-100>	staff.K
192.168.1.*	music.K
192.168.2.*	gfx.K

Each command file in the above example should contain the appropriate transcripts and consequently radmind will distribute the corresponding files to the different OS X clients in your organisation. You may use the same transcript in more than one command file.

Alternatively you can choose to use certificates to hand out command files. Certificates allow the server to identify the client and serve it the right command file regardless of its IP address. Certificates also increase the security of your radmind installation.

The various command files are stored on the server and are served to the clients on demand. When a client initiates an update, the radmind server locates the appropriate command file for that client based on its DNS name, IP address or certificate. In the process the command file is renamed on the client to the default command.K. The client then goes on to retrieve the transcripts listed in the command file if they are out-of-date.

Loadsets, base loads and overloads

A loadset is a collection of actual files and their corresponding transcript. A base loadset containing a vanilla installation of OS X can be as large as 1.5GB containing 10,000s of files. Loadsets are stored on the radmind server.

A base loadset is the initial loadset that contains a working version of OS X and optionally additional applications.

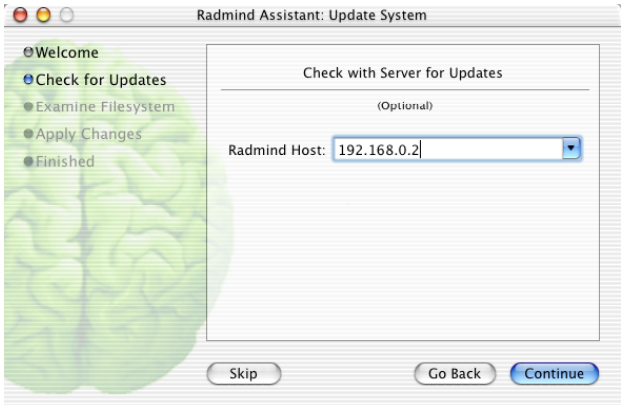
An overload is a term used to describe all loadsets other than the initial base loadset – i.e. ones that contain additional software and updates.

4. What Happens When You Update a Client?

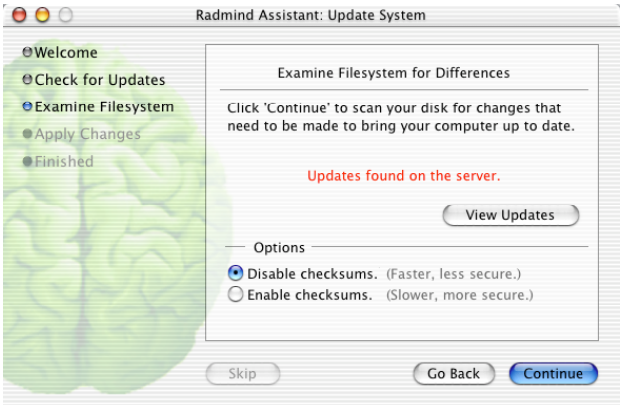
A client update consists of three stages that are reflected in the radmind Assistant. Each of these requires a user confirmation and only the last one modifies the client.

1. radmind first fetches the latest command file and associated transcripts from the server. This has no effect on the system.
2. Next radmind compares the new transcripts to the actual contents of the system and produces a temporary apply-able transcript file containing a list of required changes. This stage also has no effect on the system.
3. Finally radmind copies and deletes files on the client to match the apply-able transcript. For obvious reasons this change is normally irreversible.
4. In most cases it is a good idea to restart the computer immediately after an update. Whether this is essential or not depends on which files got replaced.

An update profoundly alters the file structure on the client and normally requires a restart. As long as you have correctly configured radmind and your negative transcript contains the Users folder, no user data will be affected. Nevertheless, having a backup is always highly recommended.



The client can access the server using its network name or IP address



This message means that the command file or transcripts on the server do not match the client. This normally means that the client is out of date.

Update Problems

If an update fails, you should not restart the computer. Instead, try to find what caused the problem. In most cases it is enough to remove the problem item from the apply-able transcript generated when the radmind Assistant checked the file system for changes to make. To do that restart the Update session, but before applying the actual loadset click on the Review Changes button and remove the offending item from the temporary transcript. Save the transcript and continue the update.

The same process can be performed from the Terminal or a script on the client:

```
ktcheck -c sha1 -h your.radmind.server
fsdiff -A / > /tmp/apply.T
lapply -h your.radmind.server
reboot
```

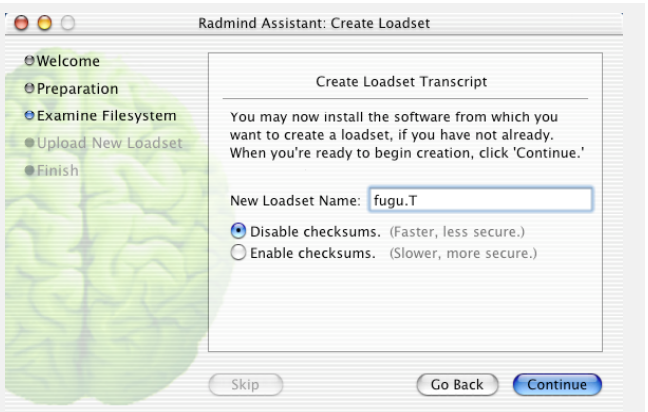
Press Command+L to see the log and learn more about the update process.

5. What Happens When You Create a New Loadset?

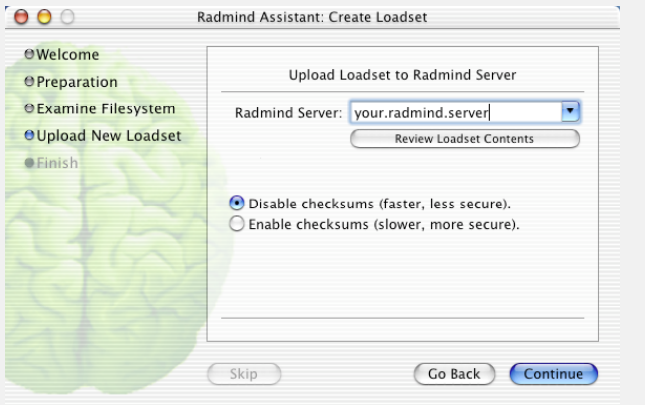
New loadsets or overloads are needed when you want to distribute new applications or files to your clients. The best method of doing this is to first update a client using radmind. This ensures that the overload correctly reflects the required changes.

1. First the client is updated using radmind. You can now install any new applications, updates, etc and then select Create New Loadset from the Session menu.
2. radmind next compares the current command file and associated transcripts against the current client configuration and produces a create-able transcript listing the contents of the new loadset you will create. You may review the transcript and possibly remove items you don't want included in the loadset by clicking the Review Loadset Contents button.
3. Once the transcript is ready it is uploaded along with the corresponding files to the designated server.

A new transcript must always be verified and deployed on the server before it is available to clients.



A transcript name helps to identify it on the server



Click the Review Loadset Contents button to open the transcript in the Transcript Editor

The same process can be performed from the Terminal or a script on the client:

```
ktcheck -c sha1 -h server.yourdomain
fsdiff -C -c sha1 / > /tmp/loadsetname.T
lcreate -h server.yourdomain /tmp/loadsetname.T
```

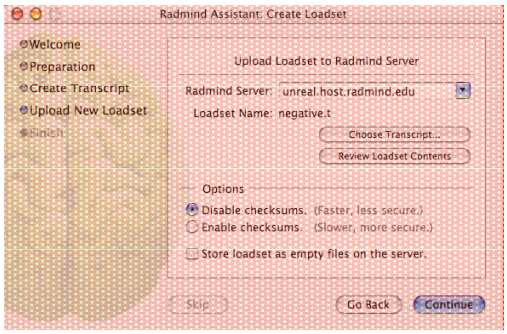
Press Command+L to see the log and learn more about the loadset creation process.

6. Managing Loadsets

It is most likely that the sample negative transcript as well as overloads you create will need editing by hand. As the number overloads and transcripts on your server grow you may also want to merge them to simplify your radmind setup. This section briefly discusses the options, but for full instructions read the Transcript Editor and Server Manager sections.

Editing transcripts

In many cases it is necessary to edit transcripts, mostly to remove lines, but in some cases you would also need to add lines to a transcript. Generally it is easier to edit transcripts on the server if you just want to delete the odd line, but if needed you can edit a transcript on a client and then upload it to the server using radmind Assistant.



The Transcript Editor is a separate application you can use to edit transcripts. Alternatively you could use a text editor such as BBEdit or vi. Make sure that you save files in Unix compatibility mode (Unix linefeeds). The default transcript editor can be set in the radmind Assistant preference panel.

On the server, Control click on a transcript to edit it. The Transcript Editor is pretty simple to use – just don't forget to save the file when done. On the server, editing a transcript is normally limited to removing items.

Adding new items to an existing negative transcript is best done on a client. First make sure you have the latest version of the transcripts by running a radmind update session. Open the transcript you want to edit in /private/var/radmind/client in the Transcript Editor and drag any files or folders you want added to the editor window. The Transcript Editor will automatically place your files in the correct sort order. Save the transcript when done.

Next you will need to upload the transcript to the server. Select Create New Loadset from the radmind Assistant Session menu and then skip the first two stages until you get to the Upload Loadset to radmind Server screen. Select your edited negative transcript and press continue. You will also need to verify and deploy the transcript on the server.

Creating overloads

With radmind any client can be used to create an overload. First update the client with radmind then install the new software. When installation is complete you can use the radmind Assistant to scan the system for changes and upload those to the server.

Another method can be helpful when you know all where the new software is placed on the system. Many OS X applications such as Microsoft Office which are installed by simply copying them to the Applications folder.



You can use the transcript editor to add items to a transcript by simply dragging them into the editor window.

If this is the case you may create the overload by opening the Transcript Editor and creating a new transcript. Drag the new folders or applications into the blank transcript window and select Add Directory and Contents in the resulting dialogue box. The Transcript Editor will automatically generate checksums and put the items in the

correct sort order. Save the file and use the Assistant to upload the transcript and files to the server.

7. radmind Assistant – Client

Session menu

Update this machine

This is the default option. Press continue to update the client. Enter IP address or name of server. Skip to continue without downloading latest command file and transcripts. This is useful if you know that you have the latest versions already on the client.

On the next screen you will see whether updates were downloaded or not. Click Continue to compare the client system to the current command file. This may take a while depending on number of installed files, hardware, etc.

The final screen lets you review the apply-able transcript before applying it on the client.

Create New radmind Loadset

This menu item starts the creation of a new loadset and uploading it to the server. The first screen reminds you that it's best to update the client before creating a new loadset. The normal procedure would be to first update the client with radmind, then install whichever application or updates you want added and then to create the new loadset.

8. The Server Manager

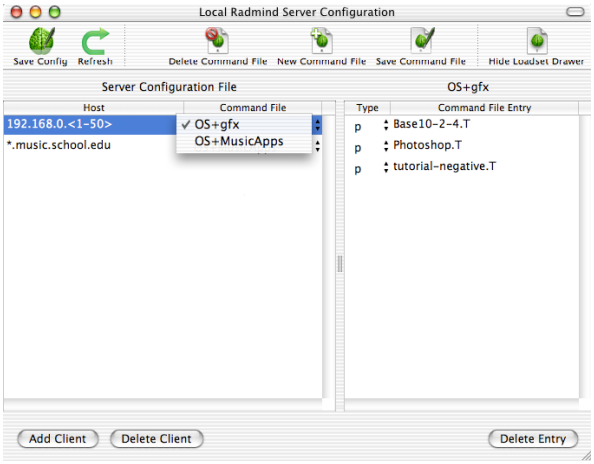
The Server Manager is only accessible if your Mac is running the server component of radmind. Select Server Manager from the Session menu to open the main window.

It is best to be logged in as root when using radmind Server Manager ???

The Server Manager window features the host (client) list on the left where each client is assigned a command file. Click on one of the hosts to reveal its command file contents on the right side of the window.

Further to the right is the transcripts pane where both newly uploaded and production transcripts are listed. Each transcript is linked to a loadset which is stored on the server.

The toolbar at the top of the window provides access to all basic functions while loadset management is done via simple drag-and-drop.



The radmind Server Manager

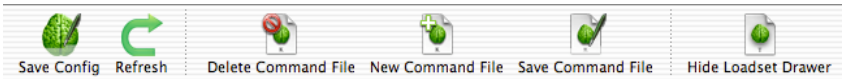
The host (client) list is displayed in the left and to its right is the command file selection menu revealing the available command files. The currently selected command file contents are shown on the right window pane.

The Loadset Drawer displays available loadsets at the top half and pending ones at the bottom.

The Interface Explained

The toolbar

The Server Manager toolbar includes buttons to create, save and delete command files as well as other key functions.



Save Config

This save the server configuration file that contains the list of clients and their assigned command files. Note that there is only one server configuration file.

Refresh

Causes the Server Manager to rescan the radmind setup folders. This is necessary in order to display newly uploaded Loadsets.

Delete Command File

Displays a dialogue box allowing you to select which command file to delete. Requires confirmation before actual deletion.

New Command File

Displays a dialogie box allowing you to name the new command file and creates it following confirmation. The new command file is then available for selection and further editing from the right window pane.

Save Command File

Saves the contents of the currently selected command file.

Hide/Show Loadset Drawer

The Loadset Drawer allows you to manage the loadsets on the server. Click this button to toggle the display of the drawer.

The server configuration

The server configuration contains the radmind client list and their associated command files. The host (client) list initially contain just one entry – the * wildcard. This enables any radmind client to connect to your server for uploads or downloads. Normally, you would want to create a client list using IP addresses or DNS names so that you can control which applications each client gets.

Click the Add Client button at the bottom of the window to create a new entry. radmind support the use of wildcards and ranges so entries such as *.music.school.edu or 192.168.1.* are valid. Click the Delete Client button to delete the current client from the list. Note that any changes must be saved using the Save Config button before they take effect.

Next to the client name or IP address is the command file popup menu used to assign a command file to a client. Selecting a client displays its current command file contents on the right window pane.

The config file is read in descending order, so you may use the wildcard at the bottom of the list to catch all undefined clients.

The command file editor

A command file can be edited by adding, removing and re-ordering items or changing their type. You can drag loadsets from the production area into the command file and you may reorder the loadsets at any time by dragging them up and down in the list.

Selection of positive, negative and special transcript type (p, n and s types) is done via the Type popup menu.

Click the Delete Entry button at the bottom of the window to remove a loadset from the command file.

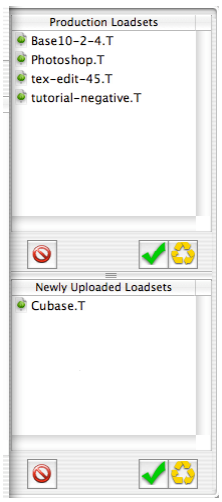
Note that any changes must be saved using the Save Command File button before they take effect.

The Loadset Drawer

The Loadset Drawer is divided in two – the bottom part displays newly uploaded loadsets while the top half shows the ones ready for deployment.

The buttons at the bottom of the drawer are used to delete, update, verify and edit the Loadsets and the same functions are also available by Control clicking a loadset. These functions can take a while depending on loadset size.

Verified loadset can be moved to the production area via drag-and-drop.



The Loadset Drawer

Delete Loadset

Deletes the selected transcript and its associated files from the server following a confirmation dialogue.

Open in Editor

Opens the appropriate transcript in the Transcript Editor. Normally, only removal of items from a loadset is practical on the server.

Verify Loadset

All newly uploaded or edited transcript s should be verified. The verification process ensures that the transcript and files in the loadset match up.

Update Loadset

This button creates or regenerates transcript checksums. It is useful if your transcript does not contain checksums or if you have manually edited one of the files in the loadset

Deploying Loadsets

When a new loadset is uploaded to the radmind server it is displayed in the bottom half of the Loadsets drawer. If you cannot see it there press the Refresh button to update the display. A loadset should be verified before adding it to the production area. Verification will fail if the transcript contains no checksums in which case use the Update (recycle) button to generate checksums before you verify.

Verified Loadsets can be dragged and dropped into the production area in the top half of the drawer to make them available for deployment.

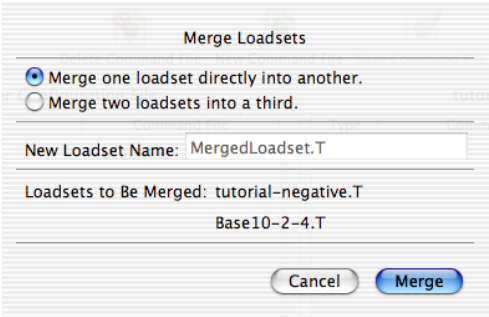
Serving a Loadset to a Client

A loadset is made available to a client by adding it to the client's command file. Select the client from the host list to display its command file contents and then drag the loadset from the drawer into the required position in the command file.

Next, select the loadset type – positive, negative or special and click Save Command File button.

Merging Loadsets

After verifying that your transcripts work correctly you may merge them to simplify your command files and to reduce storage requirements on the server. You could for example merge FileMaker601.T with FM602upd.T. To merge two Loadsets drag one on top of the other.



When merging you can either merge one transcript into another which is quicker, or to produce a third one, FileMaker602.T for example. The downside of merging in place is that you lose the ability to downgrade. The downside of producing a third transcript is that you also end up with a third loadset which increases your server storage requirements.

You may also need to edit your command files to reflect the merge. Our example command file:

```
p Base10-2-3.T
p 10-2-4update.T
p MSOffice.T
p filemaker6-02.T
p MSOffice10-1update.T
p filemaker6-03update.T
n negative.T
```

May now look like this:

```
p Base10-2-4.T
p MSOffice10-1.T
p filemaker6-03.T
n negative.T
```

9. The Transcript Editor

10. The UNIX tools

radmind is actually a collection of Unix tools that interact to achieve the complex task of client management. radmind Assistant merely serves as a front end to these tools and consequently cannot tap into all the possibilities offered by the Unix environment. From time to time you may find it useful to use the Terminal instead of the Assistant to achieve certain tasks. The full documents are available on the radmind.org site from which the following introduction is copied.

radmind - manual pages

- **ktcheck** - *command and transcript check*
  Compares local command file and associated transcripts with those stored on the radmind server. Retrieves command files or transcripts that require updates.
- **fsdiff** - *file system differences*
  Compares the file system to a set of transcripts described by a command file. Produces the differences in the form of a transcript.
- **lapply** - *loadset apply*
  Changes local machine as described by a transcript. Retrieves updated files as required from the radmind server.
- **lcreate** - *loadset create*
  Uploads a transcript and associated files to the radmind server.
- **lcksum** - *loadset checksum*
  Verifies the checksum and size of files associated with a transcript on the radmind server.
- **lfdiff** - *loadset to filesystem differences*
  Compares (with diff) a local file with the version stored on the radmind server.
- **lmerge** - *loadset merge*
  Combines two or more transcripts and corresponding files into one loadset. Works on the radmind server.
- **radmind** - *remote admin daemon*
  Implements the radmind access protocol. Makes command files, transcripts, and files available to clients.
- **twhich** - *which transcript*
  Prints the transcript associated with a file. Works on the radmind server or locally.
- **applefile** - *AppleSingle file*
  Information on the format of radmind AppleSingle Files.

You can view and edit the various radmind configuration files in a text editor of your choice. While logged in as root Select Go To Folder… in the Finder and enter /private/var/radmind to reveal the files. Alternatively use the Terminal and a text editor like VI (check http://www.vilearn.org for more info).

Appendix A – Technical Information

Network Considerations

radmind uses port 6662 by default for server-client communication. You may change the port via the command line. If your server is protected by a firewall you will need to open it to allow external access on that port. You may employ SSL encryption and specific IP ranges within radmind itself to increase your server security.

radmind also works if the client is behind a NAT/firewall – there is no need to open or re-route any ports on the client side.

With the current version of radmind (0.9.4) uploading a vanilla OS X base loadset takes an average of one hour per GB of data over 100Mbit Ethernet. Expect your base loadset to take at least 90 minutes to upload on a quiet network.

The server is used to store all the loadsets so make sure you have plenty of hard disk space on the server. It is recommended that the startup drive on this Mac has at least 20GB free space, more is highly recommended.

File Locations

The radmind executables for both client and server can be found at /usr/local/bin/. Note that at least under Jaguar (Mac OS X 10.2.x) this is not one of the default environment paths. If you don't want to have to type the full path before each radmind command enter the following line at the beginning of each radmind Terminal session:

```
setenv PATH "${PATH}:/usr/local/bin"
```

Alternatively add this line to your shell login script. If you're using tcsh (the default shell in OS X) edit the script which is found in /etc/csh.login to look something like this:

```
# System-wide .login file for csh(1).

setenv PATH "/bin:/sbin:/usr/bin:/usr/sbin:/usr/local/bin"
```

You could use radmind to manage this file and ensure that this setup is copied to all your clients.

On the Server

radmind stores its various configuration files, transcripts and loadsets in /private/var/radmind/. On the server the directory includes the following items:

- command/ – all command files are stored here
- transcripts/ – all available production transcripts
- file/ – the actual managed files that make up the various loadsets
- special/ – client specific files
- tmp/file/ – newly uploaded files before they're moved into production
- tmp/transcripts/ – newly uploaded transcripts

You can relocate the radmind server files to drives other than the startup disk by placing a symbolic link to the relocated radmind directory in /private/var/. Note that on Mac OS X you may also get to this directory using the symbolic link /var. Either way, full root access is required to browse these directories. Use sudo –s in the terminal to obtain full access.

On the client

The client only keeps one command file which is always called command.K and its associated transcripts. They can all be found at /private/var/radmind/client.

In addition, temporary items like apply-able transcripts are placed in the /tmp/ directory by radmind Assistant.

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Check List

1. chapter 4 leftovers
    - To make things simpler it is recommended that you use the latest Mac you have with a fresh installation of the latest version of OS X CD you have and nothing else. Using the latest Mac hardware means that your base loadset will be able to support all older models.
      You do not need to run any of the updates that may be available online at this stage._Alternatively, you could setup a Mac with all the latest Applications and updates and create your base loadset this way. The advantage of starting with a bare system is that you are then able to uninstall applications or downgrade OSX more easily.
    - Install radmind server (use **custom install**) on one Mac and restart. Run the radmind Assistant to configure the server.
2. Certificates
3. More CLI equivalents
    - Add single item to .T
    - Merge loadset
    - Update loadset
    - Verify loadset
    - Move to production
4. Sample negative script explained with examples, zero files
5. Menus and buttons reference for client & server, transcript editor
6. Special transcript
7. Captions for gfx
8. ihook
9. CLI in depth – Explain the CLI examples in the manual (these should mimic the assistant) then show some variations.
10. Troubleshooting
11. Incorporating new hardware
12. what exactly happens when you merge transcripts and Loadsets
13. Licensing (Adobe, FCP)
14. The various options on client during session
15. Client preferences